

MARY NJUGUNA

Decision Analytics Manager & Data Scientist

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PROFESSIONAL SUMMARY

Analytical and results-driven Decision Analytics Manager & Data Scientist with extensive expertise in credit risk modeling, scorecards, lending strategy design, and enterprise data analytics. Demonstrated track record in developing predictive risk models, business blended scoring solutions, and automated dashboards to guide institutional lending decisions. Proficient in Python, SQL, R, Power BI, and modern DevOps workflows.

TECHNICAL SKILLS

- **Programming & Databases:** Python (Pandas, NumPy, Scikit-learn, SQLAlchemy), SQL (SQL Server, SSMS), R, HTML/CSS
- **Data Science & ML:** Logistic Regression, Scorecard Development, Predictive Modeling, Time Series, Clustering (K-Means, Hierarchical), Random Forest, XGBoost
- **BI & Visualization:** Power BI, Tableau, Advanced Excel (Pivot Tables, Custom Formulas), PowerPoint
- **DevOps & Tools:** DevOps Practices, Jupyter Notebooks, Streamlit, Git & Version Control

WORK EXPERIENCE

Decision Analytics Manager

Dec 2022 – Present

Creditinfo Credit Reference Bureau Kenya Ltd | Nairobi, Kenya

- Developed custom credit risk scorecards using logistic regression to evaluate borrower creditworthiness and streamline portfolio risk assessments.
- Engineered an end-to-end business blended strategy combining company performance metrics and director risk profiles to optimize commercial credit decisioning.
- Designed data-driven agricultural lending strategies for smallholder farmers, increasing financial inclusion while maintaining strict risk controls.
- Built interactive Power BI dashboards and executive reports to translate complex credit risk metrics into actionable business strategies.
- Extracted, transformed, and queried large-scale financial datasets using Python and SQL to fulfill enterprise data and compliance requests.

Data Scientist

Jul 2022 – Oct 2022

Riset Software & Systems | Nairobi, Kenya

- Optimized enterprise data collection and ETL pipelines, cleaning and processing client inventory datasets for seamless ERP system ingestion.
- Conducted market competition analysis to identify adoption bottlenecks for ERP products, providing actionable intelligence to executive and marketing teams.
- Advised software engineering teams on data schemas and key functional features during the architectural phase of an automated hotel management system.
- Trained enterprise clients on database maintenance and troubleshooting within custom ERP environments.

Data Analyst / Software Developer

Nov 2021 – Jul 2022

Cyber Crafts Academy / Patonya Technologies | Ruiru, Kenya

- Developed and maintained web applications while designing structured instructional frameworks for web engineering and data analytical courses.
- Instructed technical cohorts in foundational programming, web administration, and data analysis packages.

KEY PROJECTS

- **Diabetic Food Recommender System:** Developed an ML-based recommendation engine using XGBoost, Random Forest, and Hierarchical Clustering to suggest low-glycemic meal options; deployed interactive web app via Streamlit.

- **Insolvency Risk Analysis:** Clustered corporate and individual financial indicators (stable, potential, at-risk) using K-Means, Hierarchical Clustering, and Random Forest classification models.
- **Customer Behavioral Segmentation:** Executed multi-region retail customer clustering using R (K-Means & Hierarchical Clustering) to enable targeted marketing strategies.
- **Energy Production Forecasting:** Applied time series forecasting models to power output datasets to project generation yield and baseline trends.

EDUCATION

Bachelor of Science in Software Engineering Murang'a University of Technology Grade: Second Class Honors (Upper Division)	Aug 2017 – Dec 2021
DevOps Engineering Certification Moringa School	Mar 2024 – Jun 2024
Data Science Professional Certificate Moringa School	Jan 2022 – Jul 2022

CERTIFICATIONS & TRAINING

- IBM Data Science Professional (Methodologies, Tools, Foundations)
- Anti-Money Laundering (AML) – CIAcademyLMS
- Sanctions Compliance & Data Protection Fundamentals – CIAcademyLMS
- Cybersecurity Awareness – CIAcademyLMS
- Communication & Teamwork Foundations – LinkedIn Learning